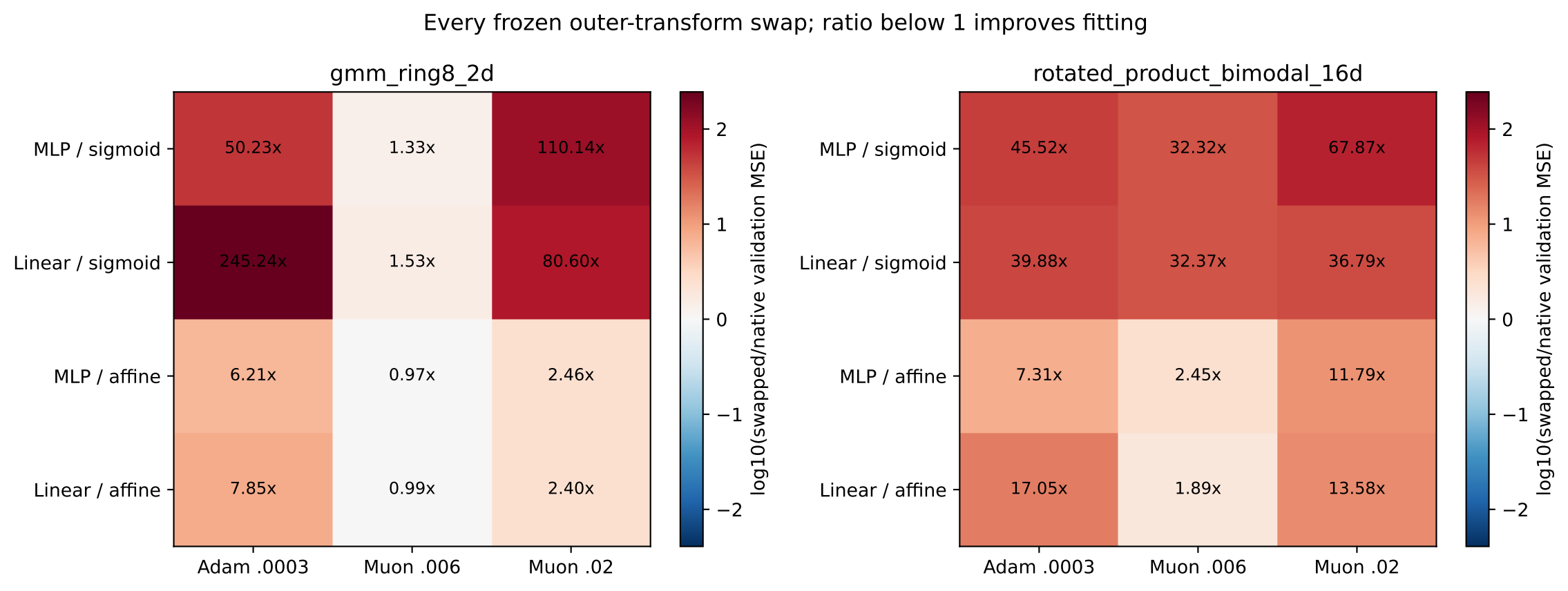
Every frozen outer-transform swap; ratio below 1 improves fitting
gmm_ring8_2d
MLP / sigmoid
Linear / sigmoid
MLP / affine
Linear / affine
50.23x
1.33x
110.14x
245.24x
1.53x
80.60x
6.21x
0.97x
2.46x
7.85x
0.99x
2.40x
Adam .0003
Muon .006
Muon .02
log10(swapped/native validation MSE)
2
1
0
−1
−2
rotated_product_bimodal_16d
MLP / sigmoid
Linear / sigmoid
MLP / affine
Linear / affine
45.52x
32.32x
67.87x
39.88x
32.37x
36.79x
7.31x
2.45x
11.79x
17.05x
1.89x
13.58x
Adam .0003
Muon .006
Muon .02
log10(swapped/native validation MSE)
2
1
0
−1
−2